



Calcium alginate–biochar composite promotes nutrient retention, enzyme activity, and plant growth in lime soil

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ABSTRACT

Soil nutrients loss in karst area is an urgent eco-environmental problem. To control soil erosion and improve soil quality, calcium alginate–rice straw biochar composite (CA–MRB300) was prepared by embedding method. The retention mechanisms of soil nutrients onto CA–MRB300 were explored through batch adsorption experiments of soil nutrients, soil column leaching experiments combined with the analysis of physicochemical properties and soil enzyme activity. The effects of CA–MRB300 and its co-application with fertilizer on plant growth were investigated by a pot experiment. The results suggested that CA–MRB300 effectively retained PO_4^{3-} and NO_3^- in lime soil mainly through surface precipitation and physical adsorption. The addition of CA–MRB300 increased the activities of alkaline phosphatase, urease, and sucrase in lime soil. Increased activity of urease accelerated the hydrolysis of urea to NH_4^+ in lime soil. Compared to a low proportion, a high proportion of CA–MRB300 was more likely to lead to the loss of K^+ . Moreover, the application of CA–MRB300 alone was more beneficial to the growth of soybean than the co-application of CA–MRB300 with fertilizer. This study indicates that CA–MRB300 has a great potential for application in retaining nutrients and promoting crop growth in lime soil.

1. Introduction

The fragile ecological environment in the karst area makes water and soil loss very easy. Under the effect of heavy rainfall and surface runoff, soil nutrients may also be lost with water, thus reducing soil fertility (Li et al., 2021b). In addition, nutrients enter rivers, lakes, and oceans with water, which would also lead to water eutrophication and adversely affect the ecological environment of karst areas (Awual, 2019; Kończak and Huber, 2022). Therefore, a green and efficient method needs to be invented to control the loss of soil nutrients. Biochar has been proved to be a green and efficient soil amendment, which can be used to improve soil fertility and promote crop growth (Dey and Mavi, 2022; Feng et al., 2021a; Wang et al., 2023). Research has also found that biochar can effectively retain cations in the soil, but its ability to retain anions is usually limited (Nan and An, 2022). Therefore, it is necessary to optimize the performance of biochar through modification. It has been proven that after ball milling, the surface characteristics of biochar can be optimized, which helps to improve the performance of biochar (Kumar et al., 2020). However, ball-milled biochar may also cause secondary pollution to the environment due to its nanoscale particle size. Therefore, it is particularly important for the large-scale

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application of biochar to improve its adsorption performance while avoiding environmental risks.

Recently, alginate has been widely used in the fields of environment and agriculture due to its good biocompatibility, non-toxic, economic, and other characteristics (Dhiman et al., 2023; Wang et al., 2019; Xiang et al., 2023). In environmental remediation, it can not only directly capture pollutants by ion exchange with pollutants, but also cross-linked with many materials to form hydrogels or as a carrier to immobilize microorganisms for pollutant removal (Wang et al., 2018a, 2018b; Zeng et al., 2022). Therefore, using alginate to synthesize biochar into calcium alginate–biochar composite (CA-BC) with larger particle sizes can simultaneously achieve the goals of improving biochar performance and avoiding environmental risks. Our previous research reported the role of CA-BC in retaining nutrients in yellow soil, but its impact on soil physicochemical properties has not been further studied (Feng et al., 2022a). In addition, different soil type may affect the nutrients retention effect of CA-BC. Therefore, it is necessary to further explore the impact of CA-BC on nutrients distribution in different karst soils.

Moreover, the activity of soil enzymes can reflect the transformation capacities of nutrients in the soil and are effective biological indicators representing soil fertility (Cui et al., 2021; Sorrenti and Toselli, 2016). Among them, soil sucrase, urease, and phosphatase can participate in the transformation of carbon (C), nitrogen (N), and phosphorus (P) in the soil (Foster et al., 2016; Noronha et al., 2022; Yuan et al., 2022). Therefore, exploring the impacts of CA-BC on soil enzyme activity is of great significance for further understanding the mechanism of nutrients retention onto CA-BC in karst soil. However, there are few reports on this aspect of research.

Based on the above reasons, calcium alginate–rice straw biochar composite (CA–MRB300) was prepared in this study, and batch adsorption experiments in lime soil, nutrients leaching experiments, and pot experiments were conducted. The objectives of this study are to 1) reveal the role of CA–MRB300 on nutrients dynamics in lime soil; 2) investigate the effects of CA–MRB300 as a soil amendment on the physiological and growth indexes of crop. This study can provide an alternative way for effectively reducing nutrients loss in lime soil.

2. Materials and methods

2.1. Preparation of materials

The lime soil was collected from the Puding Karst Ecosystem Observation and Research Station, Institute of Geochemistry, Chinese Academy of Sciences. The collection, preparation, and storage of soil samples followed the method of Bao (2000). The basic physicochemical properties (properties were shown in Table 1) of the lime soil were determined (Bao, 2000; Gaskin et al., 2008). The determination of urease, sucrase, and alkaline phosphatase activities followed the methods described by Tang et al. (2022). The lime soil pH was 7.95, electrical conductivity (EC) value was 0.172 mS cm^{-1} , and the contents of organic carbon, total N, total P, and total K were 5.63, 10.29, 0.31, and 13.65 g kg^{-1} , respectively (Table 1). In addition, urease, alkaline phosphatase, and sucrase activities of the lime soil were 0.03, 0.44, and $9.76 \text{ mg g}^{-1} 24 \text{ h}^{-1}$, respectively.

CA–MRB300 was selected as an soil amendment in this study was based on our previous research results, which demonstrated that CA–MRB300 exhibited a good retention effect on nutrients in yellow soil (Feng et al., 2022a). CA–MRB300 was prepared by sodium alginate embedding method, and the detailed preparation process followed our previous studies (Feng et al., 2022a, b). The physicochemical properties of CA–MRB300 had been discussed in previous studies, including Scanning electron microscopy/energy

Table 1
Basic properties of test soil and CA–MRB300.

Soil and biochar	Parameters	Parameter values
Lime soil	pH	7.95
	EC (mS cm^{-1})	0.172
	Organic carbon (g kg^{-1})	5.63
	Total phosphorus (g kg^{-1})	0.31
	Total nitrogen (g kg^{-1})	10.29
	Total potassium (g kg^{-1})	13.65
	Urease activity ($\text{mg g}^{-1} 24 \text{ h}^{-1}$)	0.03
	Alkaline phosphatase activity ($\text{mg g}^{-1} 24 \text{ h}^{-1}$)	0.44
	Sucrase activity ($\text{mg g}^{-1} 24 \text{ h}^{-1}$)	9.76
CA–MRB300	pH	8.75
	EC (mS cm^{-1})	0.071
	CEC (cmol kg^{-1})	154.16
	AEC (cmol kg^{-1})	9.91
	BET SSA ($\text{m}^2 \text{g}^{-1}$)	3.93
	^a Ca (g kg^{-1})	14.68
	^a Na (g kg^{-1})	1.99
	^a K (g kg^{-1})	0.32
	^a NH_4^+ (g kg^{-1})	1.95
	^a PO_4^{3-} (g kg^{-1})	0.11
	^a NO_3^- (g kg^{-1})	N.D.

Note:

^a represents the contents of extractable nutrients. N.D. represents not detected. The basic properties (excluding AEC) of CA–MRB300 presented here are derived from our previous researches (Feng et al., 2022a, b).

dispersive X-ray spectrometer (SEM-EDS, JSM-IT300, JEOL Co., Ltd.), Fourier transform infrared spectroscopy (FTIR, Nicolet iS10, Thermo Fisher Nicolet, USA), Zeta potential (Malvern Zetasizer Nano ZSE, Malvern Instruments), specific surface area (SSA, Kubo X1000, Beijing Builder Electronic Technology Co., Ltd.). The cation exchange capacity (CEC) and anion exchange capacity (AEC) of CA-MRB300 were determined according to the methods of Gaskin et al. (2008) and Zelazny et al. (1996), respectively. The contents of water-soluble nutrients in CA-MRB300 were determined, and the detailed measurement methods referred to our previous research (Feng et al., 2022a). After shaking CA-MRB300 with deionized water at a solid-liquid ratio of 1:20 (w/v) for 10 min, the pH and EC of CA-MRB300 were measured using a pH meter (PHS-3CU, Shanghai Yueping Scientific Instrument Co., Ltd.) and a soil five parameter analyzer (COMBI 5000, STEPS, German) (Rajkovich et al., 2011; Zhao et al., 2020), respectively. The basic physicochemical properties of CA-MRB300 are shown in Table 1.

2.2. Batch adsorption experiments of nutrients in lime soil

To preliminarily explore the impact of CA-MRB300 on soil nutrients (PO_4^{3-} , NO_3^- , NH_4^+ , and K^+) dynamics, batch adsorption experiments of soil nutrients were conducted. The soil sample (300 g) passing a 2 mm sieve and different proportions (0.0, 0.5, 1.5, and 3.0 wt%) of CA-MRB300 were mixed evenly in a beaker, and then moistened with deionized water. All beakers were sealed with plastic wrap and pierced with the same number of holes, and then cultured in a constant temperature incubator for 30 days (incubation temperature was 25°C, with 12 h of light and 12 h of darkness per day for a total of 30 days). After the incubation experiments were completed, the soil samples were dried and carried out the adsorption kinetics and isotherms experiments of PO_4^{3-} , NH_4^+ , NO_3^- , and K^+ in lime soil.

For the adsorption kinetics experiments, 2.0 g of incubated soil was uniformly mixed with 40 mL of ammonium phosphate solution or potassium nitrate solution (initial concentrations of PO_4^{3-} and NO_3^- were both 100 mg L⁻¹), and the mixed solutions were then oscillated at room temperature and at a speed of 150 rpm. The adsorption capacities of PO_4^{3-} , NH_4^+ , NO_3^- , and K^+ onto mixed soil were calculated at 0.083, 0.25, 0.5, 1, 2, 4, 8, 16, and 24 h, respectively. Pseudo-first-order model, pseudo-second-order model, and Elovich model were used to fit the relationships between adsorption times and adsorption capacities of PO_4^{3-} , NH_4^+ , NO_3^- , and K^+ (Wu et al., 2022a). The adsorption isotherm experiments of PO_4^{3-} , NH_4^+ , NO_3^- , and K^+ were carried out by mixing 2.0 g cultivated soil with 40 mL ammonium phosphate solution or potassium nitrate solution and shaking for 24 h. The initial concentrations of PO_4^{3-} were 0, 5, 15, 25, 50, 100, 200, and 300 mg L⁻¹, and the initial concentrations of NO_3^- were 0, 5, 10, 15, 30, 60, and 100 mg L⁻¹, respectively. The Langmuir and Freundlich models were used to fit the relationships between the initial concentrations and adsorption capacities of PO_4^{3-} , NH_4^+ , NO_3^- , and K^+ (Shi et al., 2022). To understand the adsorption mechanisms of PO_4^{3-} , NH_4^+ , NO_3^- , and K^+ by CA-MRB300 in lime soil, the adsorption isotherm data were further analyzed by site energy distribution (Cheng et al., 2021b; Yan et al., 2017).

2.3. Soil column leaching experiments

To further reveal the retention mechanisms of soil nutrients onto CA-MRB300, soil nutrient leaching experiments were conducted and soil enzyme activities were measured at different time. The detailed experimental process is as follows. A cylindrical plastic bottle with a diameter of 82 mm and a height of 245 mm was used as the experimental setup (Feng et al., 2022a). The bottom of the plastic bottle was removed, the bottle mouth was wrapped in gauze, and then was placed upside down in a 1 L glass collection bottle. The inverted bottle was filled with two pieces of filter paper, 100 g quartz sand with a diameter of 1–2 mm, 800 g mixed soil with different proportions (0.0, 0.5, 1.5, and 3.0 wt%) CA-MRB300, and one piece of filter paper from bottom to top. A total of eight treatments were set. The treatments of CA-MRB300 applied to lime soil separately in the proportions of 0.0, 0.5, 1.5, and 3.0 wt% were recorded as R0, R0.5, R1.5, and R3, respectively. Among them, R represents CA-MRB300, and the following number represents the mass percentage of CA-MRB300 applied in lime soil. In addition, the effects of co-application CA-MRB300 and fertilizer on soil nutrient loss were investigated. The treatments of co-application of different proportions of CA-MRB300 and fertilizer were labeled as NR0, NR0.5, NR1.5, and NR3, respectively, where N represents the fertilization treatment.

After the soil columns were filled, they were stabilized for three days, and then were incubated with 250 mL deionized water for seven days. 25.0 g of soil was collected to determine urease, sucrase, and alkaline phosphatase activities by the sodium phenol-sodium hypochlorite colorimetry (Dick, 2011), 3,5-dinitrosalicylic acid colorimetry (Guan, 1986), and sodium phenyl phosphate colorimetry, respectively (Dick, 2011). After soil was collected, 200 mL deionized water was slowly dropped into the soil column and recorded as the first leaching. One week later, the second leaching was carried out. In the third week, the nutrient solution prepared with urea, KCl, and KH_2PO_4 was added to the soil column to achieve N, P, and K contents of 100, 50, and 100 kg ha⁻¹, respectively (Kiani et al., 2021). After one week of incubation, 25.0 g of soil was taken out to determine soil enzyme activity. Afterward, the leaching experiment shall be carried out every seven days until the 74th day when the soil column was installed. Before the leaching, 25.0 g of soil shall be taken out to measure the enzyme activity, and then the leaching experiment shall be carried out every 14 or 35 days. The leaching experiments were conducted for 124 days and 11 times. On the 11th, 32nd, 72nd, and 124th days of the soil column leaching experiments, the urease, sucrase, and alkaline phosphatase activities of soil samples were determined (Dick, 2011; Guan, 1986). The pH, EC, and the contents of organic carbon, NH_4^+ , NO_3^- , PO_4^{3-} , K^+ , and Ca^{2+} in each filtrate were determined. The concentrations of organic carbon, PO_4^{3-} , NH_4^+ , and NO_3^- in leachate were measured on a UV-Vis spectrophotometer (T6 new century, Beijing Purkinje General Instrument Co., Ltd.) using potassium dichromate oxidation method, molybdenum antimony resistance spectrophotometry, Nessler reagent colorimetry, and ultraviolet spectrophotometry, respectively (Bao, 2000; Lu, 1999). The contents of K^+ and Ca^{2+} were analyzed using flame atomic absorption spectrophotometry on the flame atomic absorption spectrophotometer (TAS-990, Beijing Purkinje General Instrument Co., Ltd.) (Lu, 1999). At the same time, after the soil column leaching experiment, CA-MRB300 in R3 and

NR3 treatments was picked out and analyzed by X-ray photoelectron spectroscopy (XPS, K-Alpha, Thermo Scientific) and FTIR.

2.4. Pot experiments

To investigate the effectiveness of CA–MRB300 as a soil amendment, a pot experiment was conducted. Different ratios (0.0–3.0 wt %) CA–MRB300 was mixed with 800 g lime soil passing a 2 mm sieve in an automatic water-absorbing plastic flowerpot. After being stabilized for three days, the lime soil was soaked with deionized water. At the same time, in the co–application groups of CA–MRB300 and fertilizer, the same proportion of nutrient solution as the soil column experiment was added. Six soybean seeds were sown into the flowerpot, and daily management was carried out. After 30 days, the dry weight, fresh weight, stem length, root length, and chlorophyll contents of soybean were determined (Feng et al., 2022a; Hakeem et al., 2022).

2.5. Statistical analysis

The graphical analysis was carried out using SigmaPlot 14.0 and OriginPro 8.0. The site energy distribution of soil nutrients was completed using MATLAB R2018b. The Pearson’s correlation analysis was conducted to determine the correlation between soybean growth indicators, leachate properties, and soil enzyme activities using Origin 2024. The statistical analyses (one-way ANOVA) were performed to analyze the significant difference among different treatments using IBM SPSS Statistics 26.0. The least significant difference (LSD) was used to test for significance at $P < 0.05$ between the means.

3. Results and discussion

3.1. Adsorption behavior of soil nutrients onto CA–MRB300

Fig. S1 shows that the adsorption capacities of PO_4^{3-} , NH_4^+ , NO_3^- , and K^+ in lime soil with different ratios of CA–MRB300 vary with adsorption time. The adsorption capacities of PO_4^{3-} (Fig. S1a), NH_4^+ (Fig. S1b), NO_3^- (Fig. S1c), and K^+ (Fig. S1d) by different treated soils increased rapidly and then tended to equilibrium with the extension of adsorption time. Fig. S1a and Fig. S1c indicate that an increase in the application proportion of CA–MRB300 contributes to the adsorption of PO_4^{3-} and NO_3^- in lime soil, but not conducive to the adsorption of NH_4^+ and K^+ (Fig. S1b and Fig. S1d). The adsorption kinetics data of PO_4^{3-} in R0 and R0.5 were fitted with a high fitting coefficient using a pseudo–second–order model, while R1.5 and R3 more conformed to the Elovich model (Table 2). This can illustrate that if the application proportion of CA–MRB300 increases from 0.5 wt% to 1.5–3.0 wt%, the adsorption process of PO_4^{3-} can change from a chemical adsorption to the joint action of physical and chemical adsorptions (Jung et al., 2019; Zhao et al., 2023b). The adsorption kinetics data of NH_4^+ in R0, R0.5, and R1.5 were fitted with a highest fitting coefficient using a pseudo–first–order model, while the Elovich model had the highest fitting coefficient in R3. This indicates that as the application ratio of CA–MRB300 gradually increases, the adsorption process of NH_4^+ can shift from a physical adsorption to a combined effect of physical and chemical adsorption (Feng et al., 2022b). The fitting coefficient of the pseudo–first–order model for the adsorption of NO_3^- by all treatments was the highest, illustrating that the adsorption process of NO_3^- in all treatments only involved physical adsorption. The adsorption of K^+ by all treatments was fitted with three kinetic models with higher R^2 . Among them, the adsorption of K^+ in R0, R0.5, and R3 was more

Table 2
The fitting parameters of soil nutrients (PO_4^{3-} , NH_4^+ , NO_3^- , and K^+) adsorption kinetics under different application ratios (0.0, 0.5, 1.5, and 3.0 wt%) of CA–MRB300. The adsorption kinetics models include pseudo–first–order model, pseudo–second–order model, and Elovich model.

Nutrients	Biochar application rate (wt%)	Pseudo–first–order model			Pseudo–second–order model			Elovich model		
		K_1 (h ^{−1})	q_e (mg g ^{−1})	R^2	K_2 (g (mg h) ^{−1})	q_e (mg g ^{−1})	R^2	α (g (mg min) ^{−1})	β (g mg ^{−1})	R^2
PO_4^{3-}	0.0	6.8001	1.0783	0.9556	9.0438	1.1333	0.9855	1.0305	10.0503	0.9589
	0.5	6.5798	1.0934	0.9455	8.5304	1.1512	0.9836	1.0449	9.5693	0.9696
	1.5	12.0987	1.1018	0.9179	15.5076	1.1529	0.9644	1.0877	12.0919	0.9953
	3.0	13.5661	1.1553	0.9048	15.8770	1.2105	0.9570	1.1469	11.7509	0.9972
NH_4^+	0.0	14.9724	1.4873	0.9971	20.7174	1.5222	0.9967	1.4587	17.7305	0.9663
	0.5	13.3051	1.4508	0.9976	18.1181	1.4876	0.9931	1.4168	16.1290	0.9544
	1.5	16.5906	1.4463	0.9952	25.1027	1.4766	0.9948	1.4239	19.6078	0.9752
	3.0	30.9516	1.3747	0.9946	80.1732	1.3888	0.9979	1.3724	41.8410	0.9989
NO_3^-	0.0	1.5767	0.0325	0.9844	68.8752	0.0345	0.9761	1.8187	204.0816	0.9046
	0.5	4.6155	0.0542	0.9741	122.0293	0.0569	0.9373	91.5078	175.4386	0.7933
	1.5	4.3686	0.0664	0.9766	108.5920	0.0690	0.9584	295.3320	158.7302	0.8495
	3.0	4.3782	0.0850	0.9575	90.3192	0.0881	0.9616	552.8568	128.2051	0.8798
K^+	0.0	32.2344	2.6922	0.9980	50.3591	2.7132	0.9996	2.6851	28.8184	0.9988
	0.5	37.8831	2.6789	0.9990	82.4414	2.6921	0.9997	2.6748	44.2478	0.9995
	1.5	29.6812	2.5931	0.9999	50.9083	2.6095	0.9994	2.5781	39.8406	0.9958
	3.0	32.2329	2.5006	0.9973	52.6752	2.5212	0.9991	2.4952	28.5714	0.9991

Note: K_1 and K_2 are the adsorption rate constants of the pseudo-first-order and pseudo-second-order kinetic models, respectively; q_e is the adsorption capacity at equilibrium; α and β are the parameters of the Elovich model; R^2 represents the coefficient of determination (Wu et al., 2022a).

consistent with the pseudo-second-order model, while R1.5 had a higher coefficient fitted with the pseudo-first-order model. In addition, the results of adsorption kinetics indicated that adding CA-MRB300 to the soil was beneficial for the adsorption of PO_4^{3-} and NO_3^- , but not conducive to the adsorption of NH_4^+ and K^+ .

Fig. S2 reveals the change of the adsorption capacities of PO_4^{3-} (Fig. S2a), NH_4^+ (Fig. S2b), NO_3^- (Fig. S2c), and K^+ (Fig. S2d) on the lime soils treated with different proportions CA-MRB300 with the initial solution concentration. The results were consistent with those of the pure water phase, indicating that the adsorption capacities increased with the increase of initial ion concentration (Cheng et al., 2021a; Feng et al., 2022b). Fig. S2a and Fig. 2b show that PO_4^{3-} and NH_4^+ have the highest adsorption capacities in R3 and R0, respectively. The adsorption of PO_4^{3-} in lime soils of different treatments was more consistent with the Freundlich model (Table 3), indicating that the adsorption was a chemical adsorption process on heterogeneous surfaces (Guo et al., 2022). Moreover, due to the parameter $n > 1$ of the Freundlich model, PO_4^{3-} in lime soil was easily adsorbed (Wang et al., 2021b). This can also be proved by the dimensionless coefficient $K_L < 1$ of the Langmuir model. However, the adsorption of NH_4^+ , NO_3^- , and K^+ in lime soils of different treatments was more consistent with the Langmuir model. Thus, the adsorption processes of NH_4^+ , NO_3^- , and K^+ in lime soil were the uniform single-layer adsorptions that were easy to occur (Liu et al., 2021). The adsorption isotherm results further proved that CA-MRB300 was beneficial to the adsorption of PO_4^{3-} and NO_3^- , but promoted the leaching of NH_4^+ and K^+ .

3.2. Changes in pH and EC of leachates

The pH measurement results of leachates from different treatments are shown in Fig. S3a. It can be seen that the pH of leachates showed a trend of first increased and then decreased in the first 39 days, and the fluctuation range was relatively small after 46 days. The pH of different treatments during the whole experiment was 7.5–8.6. Moreover, the application of CA-MRB300 decreased the pH of leachates, which might be due to the leaching of H^+ provided by carboxyl and hydroxyl groups in CA-MRB300 during the process of simulating rainfall (Feng et al., 2022b). This result was similar to our previous results in yellow soil (Feng et al., 2022a). The changes of EC of leachates from different treatments over time are shown in Fig. S3b. The results declared that the impact trends of CA-MRB300 alone or co-application with nutrient solution on EC of leachates were similar. As the number of leaching increased, the EC of different treatments decreased gradually, and changed slightly after the fifth leaching. This indicates that Ca, Mg, and other mineral elements are extensively leached under the action of the first simulated rainfall (Liang et al., 2006). This is consistent with the results in yellow soil (Feng et al., 2022a). The increase in the application proportion of CA-MRB300 led to an increase in the EC of leachates. The EC of R3 was 15.47 times that of R0 at the first leaching. Research has shown that CA-MRB300 contains many Ca^{2+} (Feng et al., 2022a). Therefore, CA-MRB300 increased the EC of leachate because CA-MRB300 released Ca^{2+} , and the first leaching released the most Ca^{2+} .

3.3. Changes of nutrients content in leachates

Fig. 1 represents the cumulative leaching concentration of nutrients in leachates. After eight times of simulated rainfall, PO_4^{3-} of different treatments almost did not leach (Fig. 1a). Significantly, CA-MRB300 can reduce the loss of PO_4^{3-} in lime soil, and the higher the proportion of CA-MRB300 in the soil column, the lower the leaching amount of PO_4^{3-} . When CA-MRB300 was applied separately, the cumulative leaching amount of PO_4^{3-} in R0.5, R1.5, and R3 decreased by 10.92 %, 25.14 %, and 28.58 % compared with R0,

Table 3

The fitting parameters of soil nutrients (PO_4^{3-} , NH_4^+ , NO_3^- , and K^+) adsorption isotherms under different application ratios (0.0, 0.5, 1.5, and 3.0 wt%) of CA-MRB300. The adsorption kinetics models include Langmuir model and Freundlich model.

Nutrients	Biochar application rate (wt%)	Langmuir model			Freundlich model		
		K_L (L mg ⁻¹)	Q_m (mg g ⁻¹)	R^2	K_F (L g ⁻¹)	n	R^2
PO_4^{3-}	0.0	0.0473	2.3996	0.9638	0.3443	2.7352	0.9853
	0.5	0.0400	2.6625	0.9566	0.3948	2.8257	0.9947
	1.5	0.0385	2.9455	0.9525	0.4462	2.8645	0.9923
	3.0	0.0451	3.0576	0.9562	0.4954	2.9138	0.9947
NH_4^+	0.0	0.0055	5.0909	0.9989	0.0906	1.5921	0.9909
	0.5	0.0057	4.7544	0.9985	0.0902	1.6121	0.9890
	1.5	0.0063	4.2857	0.9989	0.0963	1.6801	0.9909
	3.0	0.0073	3.7123	0.9974	0.1023	1.7627	0.9886
NO_3^-	0.0	0.1371	0.0394	0.9931	0.0134	4.4209	0.9802
	0.5	0.1344	0.0543	0.9972	0.0182	4.3764	0.9693
	1.5	0.1424	0.0751	0.9994	0.0262	4.4053	0.9087
	3.0	0.1648	0.0910	0.9997	0.0337	4.7326	0.9702
K^+	0.0	0.0160	3.5438	0.9997	0.1532	1.7640	0.9886
	0.5	0.0163	3.1902	0.9977	0.1456	1.8051	0.9750
	1.5	0.0147	3.1497	0.9974	0.1289	1.7612	0.9787
	3.0	0.0127	3.0079	0.9996	0.1064	1.7094	0.9874

Note: Q_m is the maximum adsorption capacity; K_L is a parameter that Langmuir characterizes the affinity between the adsorbent and the adsorbate; K_F is the Freundlich adsorption capacity parameter; n describes the Freundlich isotherm change trend; R^2 represents the coefficient of determination (Shi et al., 2022).

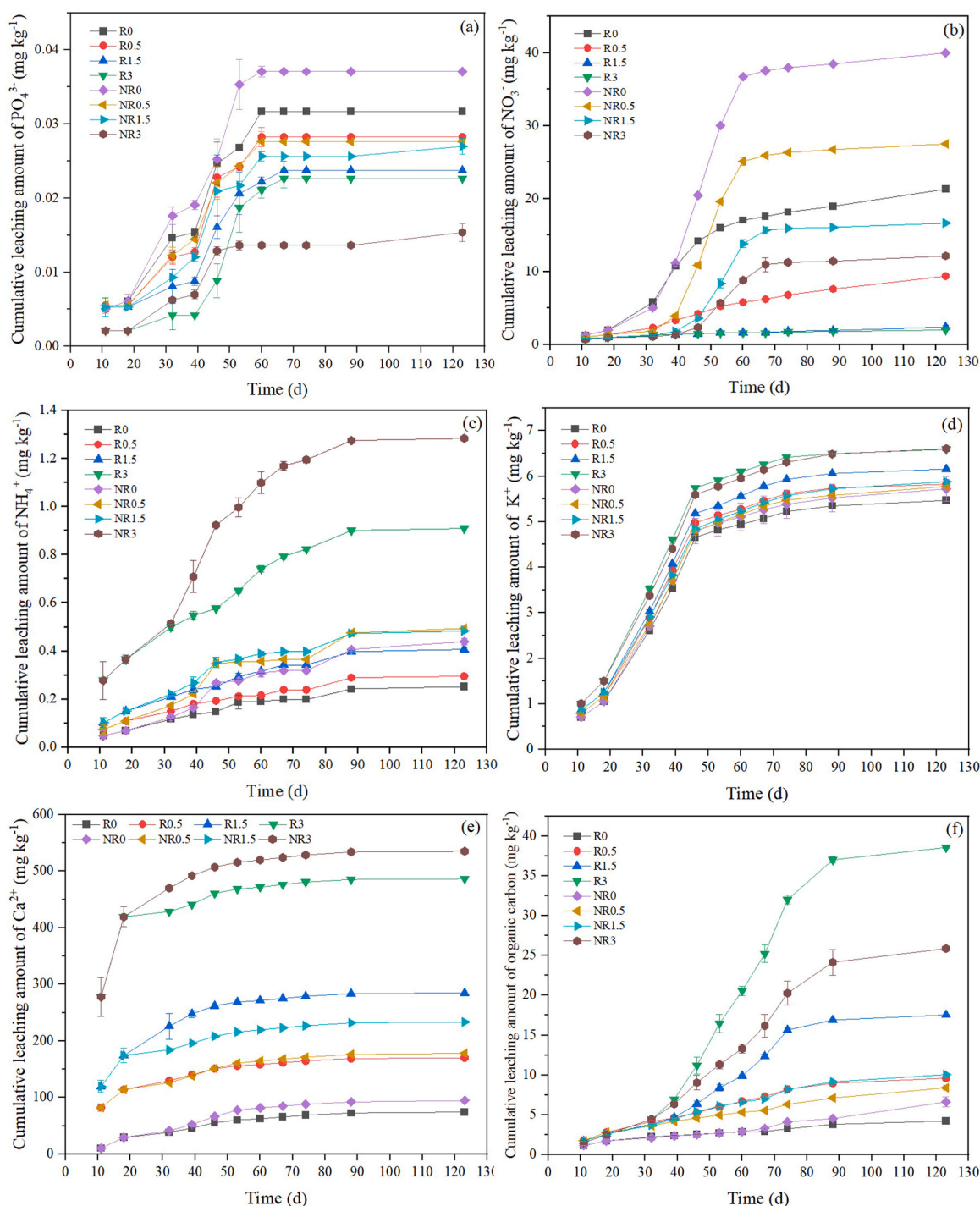


Fig. 1. Effects of different treatments on accumulative PO_4^{3-} (a), NO_3^- (b), NH_4^+ (c), K^+ (d), Ca^{2+} (e), and organic carbon (f) contents of leachates. R0, R0.5, R1.5, and R3 represent the application ratios of CA-MRB300 in lime soil at 0.0, 0.5, 1.5, and 3.0 wt%, respectively. NR0, NR0.5, NR1.5, and NR3 indicate the co-application of fertilizer and different ratios of CA-MRB300 in lime soil.

respectively. When CA-MRB300 was applied with nutrient solution, the cumulative leaching amount of PO_4^{3-} in NR3 was 57.26 % less than that in NR0. CA-MRB300 can reduce the loss of PO_4^{3-} in lime soil, which may be due to PO_4^{3-} being retained through ligand exchange or precipitation formation with Ca^{2+} contained in CA-MRB300 (Mia et al., 2017). Moreover, the addition of CA-MRB300 can increase the content of soil organic carbon, which can provide active sites for the retention of PO_4^{3-} in the soil, thereby reducing the leaching amount of PO_4^{3-} (Xu et al., 2018b). Other studies have also found that biochar has the potential to enhance PO_4^{3-} retention in

soil (Purkayastha et al., 2022; Wu et al., 2021).

Compared to PO_4^{3-} , the application of CA-MRB300 had a better effect in controlling the loss of NO_3^- in lime soil (Fig. 1b). After 11 times of leaching, the cumulative loss of NO_3^- in R0.5, R1.5, and R3 reduced by 56.06 %, 88.79 %, and 90.71 % respectively compared with R0, while reduced by 31.24 %, 58.36 %, and 69.64 %, respectively in NR0.5, NR1.5, and NR3 compared with NR0. Therefore, the application of 3.0 wt% CA-MRB300 alone was more beneficial to control the loss of NO_3^- in lime soil than that of the co-application with nutrient solution. The role of CA-MRB300 on NO_3^- in lime soil was similar to that of mesquite biochar and apple branch biochar (Beusch et al., 2019; Cao et al., 2019). According to previous researches, CA-MRB300 has a certain water retention ability due to the presence of functional groups such as phenols and alcohols (Cao et al., 2019; Feng et al., 2022a; Zhao et al., 2023a). The hydrated asymmetric NO_3^- flowing into CA-MRB300 with the solution can be captured in the pores of CA-MRB300 through physical adsorption (Li et al., 2018). The ion exchange capacity of CA-MRB300 and the reaction of its surface functional groups with NO_3^- may also help reduce the loss of NO_3^- (Kameyama et al., 2012; Li et al., 2018). In addition, research has also found that biochar contains certain reducing functional groups (such as quinone functional groups), which can reduce nitrate as an electron donor to NH_4^+ (Saquing et al., 2016). To sum up, CA-MRB300 had a good effect on the retention of NO_3^- in lime soil.

As shown in Fig. 1c, the application of CA-MRB300 had a positive impact on the leaching of NH_4^+ in lime soil. After the experiments, the cumulative losses of NH_4^+ in R3 and NR3 were 72.25 % and 65.77 % higher than that from R0 and NR0, respectively. This may be because CA-MRB300 promotes the activity of soil enzymes, inducing the hydrolysis of urea to NH_4^+ , which are leached out under the action of water (Liu et al., 2018). Moreover, after adding CA-MRB300, the NO_3^- content in the leachate decreased (Fig. 1b), thus the increase of NH_4^+ concentration might also be caused by CA-MRB300 induced NO_3^- reduction to NH_4^+ (Saquing et al., 2016).

The application of CA-MRB300 promoted the leaching of K^+ in lime soil, and the high proportion of CA-MRB300 (3.0 wt%) had a great contribution to K^+ leaching (Fig. 1d). Compared with R0 and NR0, after 11 times of simulated rainfall events, the cumulative losses of K^+ in R3 and NR3 increased by 13.37 %-17.02 %. The research results of Yuan et al. (2016) also showed that K^+ contained in

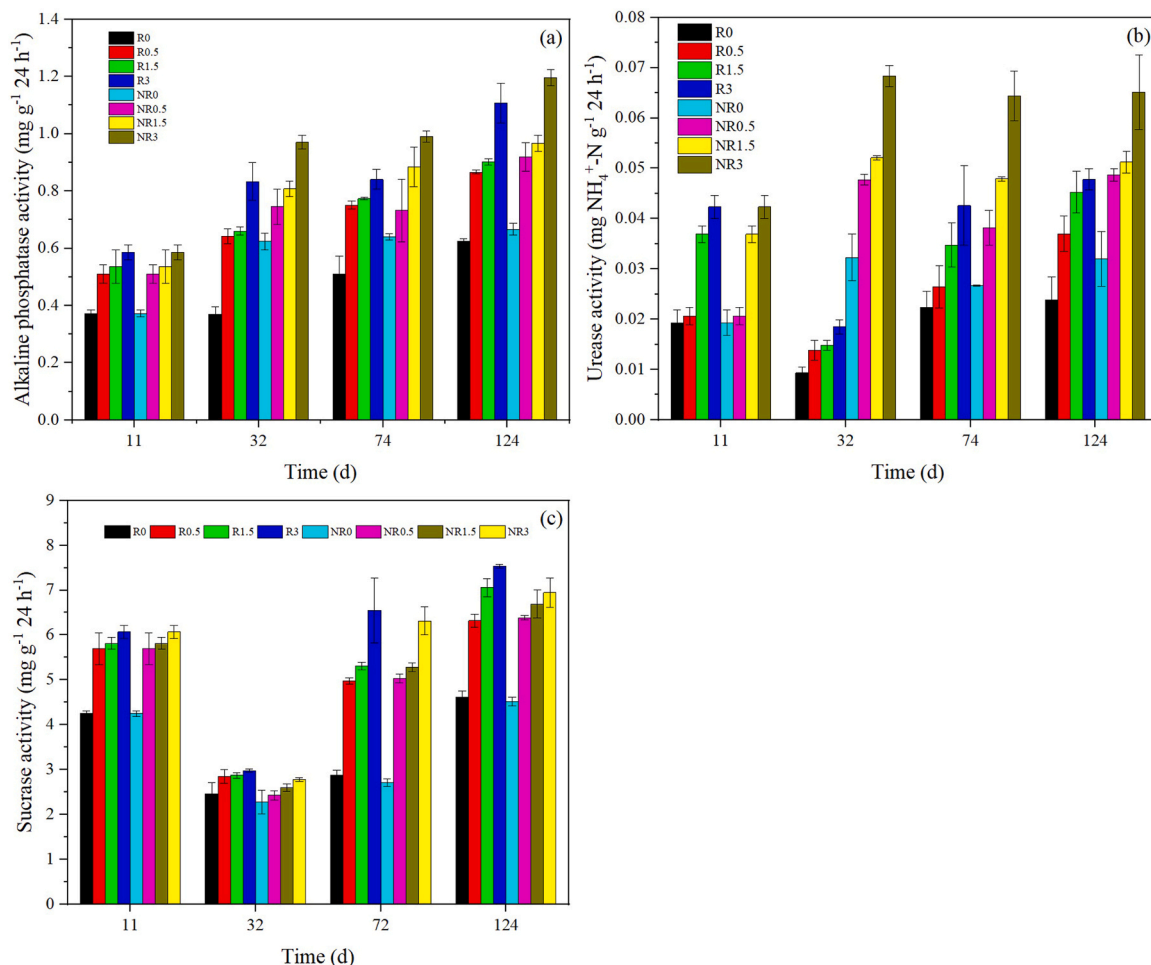


Fig. 2. Effects of different treatments on soil alkaline phosphatase (a), urease (b), and sucrase (c). R0, R0.5, R1.5, and R3 represent the application ratios of CA-MRB300 in lime soil at 0.0, 0.5, 1.5, and 3.0 wt%, respectively. NR0, NR0.5, NR1.5, and NR3 indicate the co-application of fertilizer and different ratios of CA-MRB300 in lime soil.

biochar could be released from soil column experiments, and compared to the treatment without biochar, the concentration of K^+ in leachate after adding biochar prepared by pyrolysis at 300 °C and 500 °C increased. Since CA–MRB300 itself contained 320 mg kg^{-1} of K^+ (Table 1), thus the increase of K^+ concentration in leachate might be due to the release of K^+ from the composite itself.

To further explain the above results, the concentrations of Ca^{2+} (Fig. 1e) and organic carbon (Fig. 1f) in leachates collected at different times were measured. The application of the composite significantly increased the concentration of Ca^{2+} in leachates, but the cumulative concentration of Ca^{2+} basically reached equilibrium on the 60th day, which was similar to the changing trend of EC of leachates. After the experiments, the cumulative concentration of Ca^{2+} in the leachates of R0.5, R1.5, and R3 increased by 56.33 %, 73.88 %, and 84.71 %, respectively compared with R0, and the cumulative leaching concentration of Ca^{2+} of NR0.5, NR1.5, and NR3 increased by 46.77 %, 59.39 % and 82.29 % respectively compared with NR0. Previous studies declared that the concentration of Ca^{2+} in the extract of CA–MRB300 was 14.68 g kg^{-1} (Feng et al., 2022a), and these Ca^{2+} were evenly distributed on the surface of CA–MRB300, which could be leached with water. The large amount of Ca^{2+} contained in CA–MRB300 may have a significant impact on the fate of soil nutrients through physical or chemical processes. For example, PO_4^{3-} in the soil can form precipitation with Ca^{2+} contained in CA–MRB300 to reduce the leaching of PO_4^{3-} (Mia et al., 2017), which may be an effective way to reduce the loss of PO_4^{3-} in lime soil.

It can be seen that the concentrations of organic carbon in leachates were in direct proportion to the application dose of the composite in lime soil (Fig. 1f). After 11 times of leaching, the organic carbon contents of leachates in R3 and NR3 were 89.05 % and 74.41 % higher than that of R0 and NR0, respectively. The results of Feng et al. (2021b) also reveals that wheat straw and cow dung biochars improve the concentration of organic matter in loamy sandy soil by releasing organic matter (Feng et al., 2021b). These findings further confirm that the application of CA–MRB300 can contribute to a certain soil organic matter pool, thereby providing active sites for the adsorption of soil nutrients (such as PO_4^{3-}) (Xu et al., 2018b).

In conclusion, CA–MRB300 decreased the loss of PO_4^{3-} and NO_3^- in lime soil mainly through surface precipitation and physical adsorption. The increase of NH_4^+ leaching by CA–MRB300 mainly because the increase of soil urease activity accelerated the hydrolysis of urea to NH_4^+ . In addition, CA–MRB300 increased the loss of K^+ might be due to the leaching of K^+ from the composite itself.

3.4. Soil enzyme activity

The effects of different treatments on soil alkaline phosphatase, urease, and sucrase are shown in Fig. 2. The addition of CA–MRB300 promoted the activity of soil alkaline phosphatase (Fig. 2a). Compared with R0 and NR0, at 124 days, R3 and NR3 increased by 43.6 % and 44.3 %, respectively. The soil alkaline phosphatase activity was higher when CA–MRB300 was applied with fertilizer than CA–MRB300 was applied alone. This may be because the co-application of CA–MRB300 and fertilizer provides nutrients to the lime soil, increasing the alkaline phosphatase activity in lime soil. The same phenomenon has also been observed in other studies (Azeem et al., 2019; Nahidan and Ghasemzadeh, 2022). These results demonstrate that biochar is feasible in increasing soil alkaline phosphatase activity (Jin et al., 2016).

Fig. 2b reveals the change of soil urease activity under different treatments. It showed that the addition of CA–MRB300 promoted soil urease activity, and R3 and NR3 had the best promotion effect on soil urease activity. Urease has the function of hydrolyzing organic compounds (urea), thus increased urease activity in lime soil indicates that the hydrolysis of urea is enhanced (Lopes et al., 2021). In addition, urease mediated urea hydrolysis is usually accompanied by the production of NH_4^+ , thus increased urease activity in R3 and NR3 indicates that the concentration of NH_4^+ also increases in these two treatments (Liu et al., 2018). As shown in Fig. 1c, CA–MRB300 can lead to the loss of soil NH_4^+ , and the concentration of NH_4^+ usually has a certain correlation with urease activity. Therefore, increased urease activity promotes the hydrolysis of urea to NH_4^+ . When the amount of NH_4^+ produced is greater than the amount of adsorption, the leaching of NH_4^+ increases (Liu et al., 2018). In addition, compared with single application, there was a greater increase in soil urease activity when the co-application of CA–MRB300 combined with nutrient solution. This is because the addition of nutrient solution decreases the C/N ratio of the soil, thereby increasing the hydrolysis of urea (Lopes et al., 2021).

Fig. 2c presents that CA–MRB300 is beneficial to the increase of soil sucrase activity. Large particle size CA–MRB300 can improve soil permeability and provide a carbon source, thus improving the living environment of microorganisms (Wang et al., 2022). Throughout the whole experiment process, sucrase activity was first inhibited and then promoted. On the 32nd day, significant inhibition of activity may be attributed to a decrease in soil moisture during the incubation process (Brockett et al., 2012; Xing et al., 2022). The results also illuminated that the presence or absence of nutrient solution had little effect on sucrase activity. Previous research also suggested that adding biochar increased sucrase activity in the soil because it could provide organic carbon for the soil (Cui et al., 2019). The sucrase activity is usually related to the content of soil organic matter, and the increase of activity can enhance the decomposition and transformation of organic carbon in the soil (Cui et al., 2019). Our research results also showed that the application of CA–MRB300 can significantly increase the content of organic carbon in the leachates, especially in the treatments with the addition of 3 wt% CA–MRB300 (Fig. 1f). Therefore, the increase in soil sucrase activity may be due to the abundant organic carbon concentration provided by CA–MRB300 to the soil. In a word, CA–MRB300 promoted the activities of alkaline phosphatase, urease, and sucrase in lime soil, and the co-application of CA–MRB300 and nutrient solution had a good promoting effect.

3.5. Retention mechanism of soil nutrients by CA–MRB300

Fig. S4 represents the XPS orbital peaks of C 1 s, O 1 s, Ca 2p, N 1 s, and P 2p of CA–MRB300 before and after soil column leaching. The C 1 s peak was divided into C–C (284.5–284.7 eV), C–O (285.4–286.1 eV), C=O (286.5–286.8 eV), and O=C–O (288.3–288.7 eV), respectively (Fig. S4a) (Alhujailey et al., 2020). C–C occupied the leading position of CA–MRB300, and the ratio of

C–C and O=C–O decreased after leaching, but the ratio of C–O increased. The relative proportion of O=C–O in CA–MRB300 decreased from 14.12 % of R (fresh CA–MRB300) to 10.78 % of R3 and 10.59 % of NR3, which might be caused by the ion exchange of NH_4^+ on $-\text{COOH}$ (Wu et al., 2022b). The O 1s peak was divided into five peaks, representing O–C (530.5–531.5 eV), C–O–Ca (532.0–532.2 eV), C–O–C (532.7–533.0 eV), O=C (533.4–533.8 eV), and O–H (531.1–531.3 eV), respectively (Fig. S4b) (Wang et al., 2021a). C–O–Ca, C–O–C, and O–H were dominant, and C–O–Ca and O–H in R3 and NR3 were lower than R, suggesting that O–H and C–O–Ca participated in nutrients retention. The binding sites of O–H and Ca can form complexes with H_2PO_4^- to adsorb H_2PO_4^- (Eduah et al., 2020). The Ca 2p peak was deconvoluted into three peaks (Fig. S4c), representing Ca $2p_{3/2}$ (347.6–349.4 eV), Ca $2p_{1/2}$ (350.9–351.9 eV), and CaO (346.9–348.1 eV) (Ji et al., 2022). Compared with R, the contents of Ca $2p_{3/2}$ in R3 and NR3 decreased, but Ca $2p_{1/2}$ increased, indicating that Ca^{2+} and NH_4^+ were adsorbed through ion exchange. This can be proved by the increase of the relative contents of NH_4^+ in R3 and NR3 in the N 1s orbital peak. The N 1s peak was divided into three peaks, representing C=N (400.1–401.3 eV), N–(C=O) (403.3–407.9 eV), and NH_4^+ (399.4–400.5 eV), respectively (Fig. S4d) (Wang et al., 2021a). Similarly, the P 2p peak was also divided into three peaks, representing HPO_4^{2-} (133.4–135.3 eV), PO_4^{3-} (126.5–135.3 eV), and H_2PO_4^- (138.8–140.7 eV), respectively (Fig. S4e) (Xu et al., 2018a). Among them, H_2PO_4^- occupied a dominant position. The soil pH was slightly alkaline, thus the main forms of phosphate were HPO_4^{2-} and H_2PO_4^- (Pan et al., 2009). The relative contents of HPO_4^{2-} had a certain change before and after leaching, this might be due to the formation of surface precipitation with Ca^{2+} (Li et al., 2021a). The peaks of C 1s, Ca 2p, N 1s, O 1s, and P 2p were respectively represented at 284.53, 347.36, 399.53, 532.43, and 132.86 eV (Fig. S4f). The peak Ca 2p of the non-leached CA–MRB300 was more obvious compared with the leached CA–MRB300, indicating that the Ca in the CA–MRB300 had reacted in the soil. The N 1s peak of R3 and NR3 was larger than that of R, indicating that CA–MRB300 had successfully retained N in the soil. However, the peak of P 2p had not changed much, this may be because the concentration of PO_4^{3-} in the soil was less than that of NO_3^- and K^+ , thus PO_4^{3-} in the soil was less than that of the other two nutrients.

The FTIR analysis diagrams before and after CA–MRB300 leaching are shown in Fig. S5. The result showed that the types and contents of surface functional groups in CA–MRB300 had changed before and after leaching. The peaks at 3447–3463 and 1630–1651 cm^{-1} represent the O–H stretching vibration peak and C=O peak, respectively (Zhang et al., 2015; Zhao et al., 2020), and the peaks at 1397–1432, 1030–1097, and 469 cm^{-1} represent the stretching vibration peak of C=C, C–O, and C–C, respectively (Feng et al., 2022b; Zornoza et al., 2016). It can be seen from Fig. S5 that fresh CA–MRB300, the leached soil and CA–MRB300 contained relatively rich O–H, C=C, C–O, and C–C, which may play an important role in the dynamics of soil nutrients. For example,

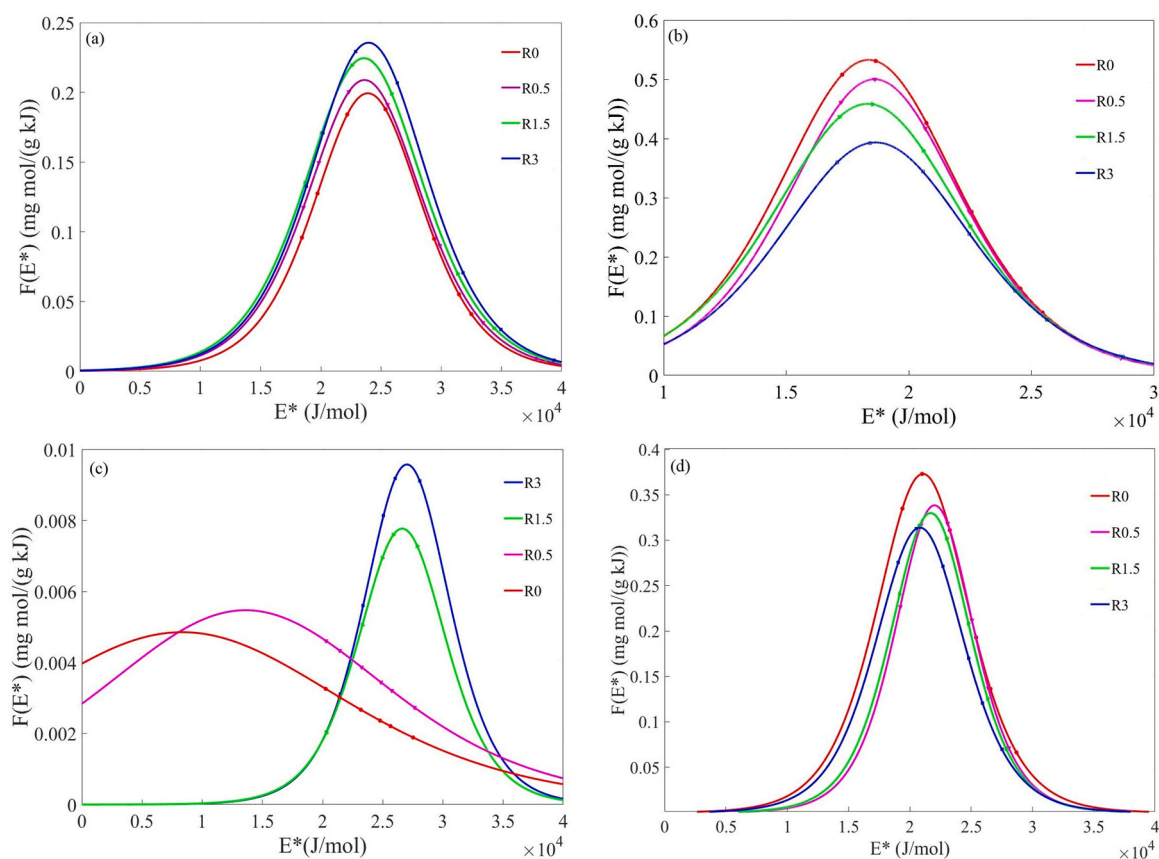


Fig. 3. Site energy distribution of soil PO_4^{3-} (a), NH_4^+ (b), NO_3^- (c), and K^+ (d) in different treatments. R0, R0.5, R1.5, and R3 represent the application ratios of CA–MRB300 in lime soil at 0.0, 0.5, 1.5, and 3.0 wt%, respectively.

-OH on the surface of CA-MRB300 can undergo ligand exchange with H_2PO_4^- in soil, thereby reducing nutrient loss (Mia et al., 2017).

To further understand the main adsorption mechanisms of soil nutrients, site energy analysis was conducted (Fig. 3). The E^* of the four nutrients was different, indicating that the adsorption process of the four nutrients might be different (Li et al., 2021a). The difference of E^* values of R0, R0.5, R1.5, and R3 was small (Fig. 3a), explaining that different ratios of CA-MRB300 had a relatively small impact on the adsorption process of PO_4^{3-} . However, the high addition ratio of CA-MRB300 was helpful for the increase of the $F(E^*)$. In R3, the high adsorption energy sites were preferentially occupied by PO_4^{3-} , resulting in the increased adsorption of PO_4^{3-} (Zhang et al., 2022). The adsorption of NH_4^+ was opposite to that of PO_4^{3-} . The decrease in the addition ratio of CA-MRB300 had an adverse effect on the adsorption of NH_4^+ in lime soil. For NO_3^- , R0, R0.5, R1.5, and R3 showed different results from NH_4^+ . There was a positive correlation between the increase in CA-MRB300 addition amount and the increase in $F(E^*)$ value (Fig. 3c), demonstrating that CA-MRB300 promoted the adsorption of NO_3^- . The E^* values of R0-R0.5 and R1.5-R3 were different, suggesting that the adsorption process of NO_3^- in R0, R0.5, R1.5, and R3 might be different. The adsorption of K^+ in R0, R0.5, R1.5, and R3 was similar to that of NH_4^+ . The $F(E^*)$ values gradually decreased with increased CA-MRB300, suggesting that CA-MRB300 inhibited the adsorption of K^+ in lime soil (Fig. 3d). The adsorption energy E^* of PO_4^{3-} adsorbed by different treated soils was 2.24–2.54 kJ mol^{-1} , manifesting that physical and chemical action dominated the adsorption of PO_4^{3-} . The results of XPS revealed that the adsorption mechanism of PO_4^{3-} in R3 and NR3 was surface precipitation (Li et al., 2021a). The adsorption energy E^* of different treated soils for NH_4^+ adsorption was 1.84–1.86 kJ mol^{-1} , indicating that the adsorption of NH_4^+ was mainly physical process. The adsorption energy E^* of different treated soils for NO_3^- and K^+ adsorption was 2.03–2.60 and 2.05–2.30 kJ mol^{-1} , respectively, indicating that physicochemical adsorption dominated the adsorption processes of NO_3^- and K^+ . In conclusion, the adsorption mechanism of PO_4^{3-} by CA-MRB300 in lime soil was mainly surface precipitation, while for NO_3^- was mainly physical adsorption.

3.6. Effect of CA-MRB300 as a soil amendment on soybean growth

The growth of soybean in each treatment is shown in Fig. S6. It can be seen that the soybean growth of each treatment was good in the experimental process. Compared to the treatments without adding CA-MRB300, the treatments with CA-MRB300 resulted in a more developed soybean root system (Fig. 4). In addition, the phenomenon of CA-MRB300 applied to the soil being penetrated by soybean roots was also observed (at the black dots in the Fig. 4), indicating that CA-MRB300 had a positive impact on the development of soybean roots. This may be because CA-MRB300 can provide nutrients for soybean (Feng et al., 2022a), which is beneficial for their growth.

Fig. 5 reveals the effects of different treatments on fresh weight, dry weight, stem length, and chlorophyll content of soybean. The results proclaimed that R0.5, R1.5, and R3 increased the fresh weight of soybean, but the difference with R0 was not significant (Fig. 5a). The application of CA-MRB300 alone could promote the growth of soybean more than co-application of CA-MRB300 and fertilizer (especially NR3). On the one hand, CA-MRB300 could release Ca^{2+} , and the content released into the soil increased as the added proportion of CA-MRB300 increased (Fig. 1e). On the other hand, the content of K^+ in lime soil could be increased after CA-MRB300 was applied with fertilizer. Therefore, when a high proportion of CA-MRB300 was co-applied with fertilizer, soybeans may be subjected to salt stress due to high content of soluble salt in the soil, which was detrimental to the growth of soybean (Karabay et al., 2021). Fig. 5b shows the dry weight of different tissues in soybean. The dry weight of aboveground parts was significantly higher

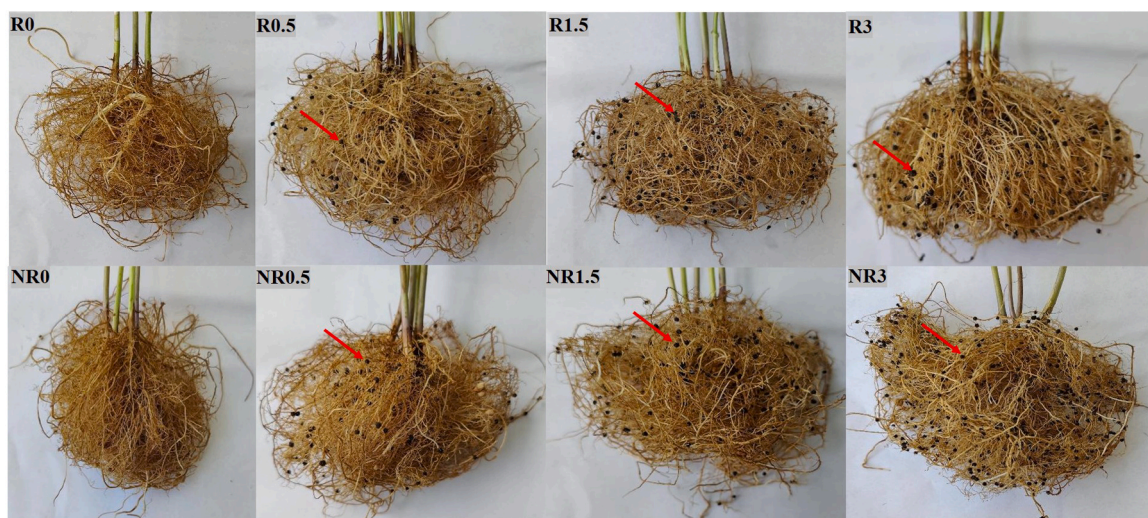


Fig. 4. Effects of different treatments on soybean root growth. R0, R0.5, R1.5, and R3 represent the application ratios of CA-MRB300 in lime soil at 0.0, 0.5, 1.5, and 3.0 wt%, respectively. NR0, NR0.5, NR1.5, and NR3 indicate the co-application of fertilizer and different ratios of CA-MRB300 in lime soil. The black ball pointed by the red arrow is the CA-MRB300 that has been penetrated by the soybean root.

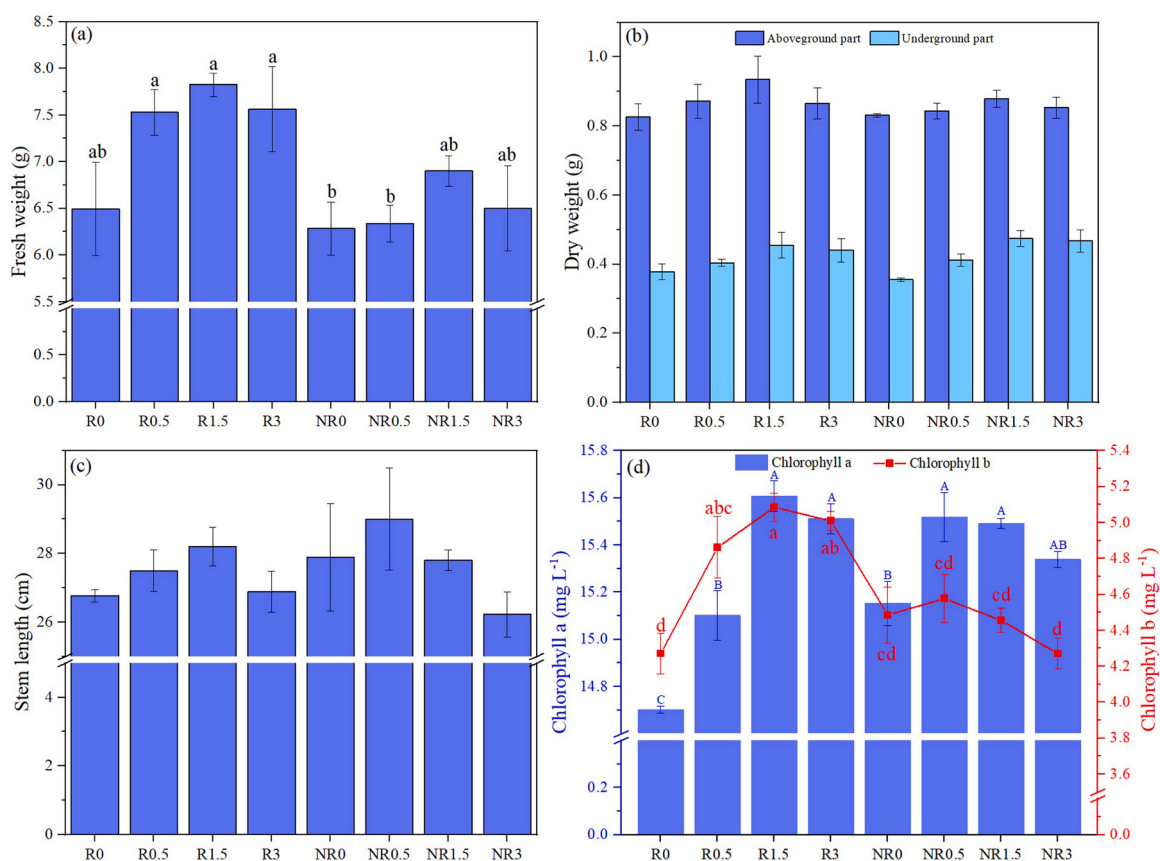


Fig. 5. Effects of different treatments on fresh weight (a), dry weight (b), stem length (c), and the contents of chlorophyll a (left Y-axis) and chlorophyll b (right Y-axis) (d) of soybean. R0, R0.5, R1.5, and R3 represent the application ratios of CA–MRB300 in lime soil at 0.0, 0.5, 1.5, and 3.0 wt%, respectively. NR0, NR0.5, NR1.5, and NR3 indicate the co-application of fertilizer and different ratios of CA–MRB300 in lime soil. Data are means $\pm$ SD ($n=3$). Different lowercase or uppercase letters represent statistical differences between different treatments ($P<0.05$). The absence of letters in Fig. 5b and c indicates no significant difference between different treatments. The split axes and bars indicate that the “removal” of axis scale lines and data graphs from the starting point to the ending point of the breakpoint by setting breakpoints during the drawing process.

than that of underground parts. The difference in stem length between different treatments was not significant (Fig. 5c). The application of CA–MRB300 promoted the contents of soybean chlorophyll a, but the promotion effect of 3 wt% CA–MRB300 on soybean chlorophyll decreased (Fig. 5d). In general, the application of 1.5 wt% CA–MRB300 alone was most beneficial to the growth of soybean.

3.7. Correlation between the growth of soybeans and the properties of leachate and soil

The correlation analysis was used to determine the correlation between soybean growth indicators and the properties of leachate and soil. As shown in Fig. 6, the dry weight of soybeans was significantly positively correlated with K^+ and Ca^{2+} contents in leachate and alkaline phosphatase activity in soil ($P<0.05$). The chlorophyll content of soybeans was significantly positively correlated with the contents of K^+ and organic carbon in leachate and alkaline phosphatase activity in soil ($P<0.05$). Meanwhile, the dry weight and chlorophyll content were significantly positively correlated with soil sucrase activity ($P<0.01$). There was a significant positive correlation between fresh weight and soil sucrase activity ($P<0.05$). These results indicated that after adding CA–MRB300, K^+ , Ca^{2+} , organic carbon contents, alkaline phosphatase and sucrase activities in lime soil can directly affect the growth of soybeans. Moreover, alkaline phosphatase activity was significantly positively correlated with K^+ , Ca^{2+} , NH_4^+ , organic carbon, EC, sucrase activity, and urease activity ($P<0.01$). Sucrase activity was significantly positively correlated with K^+ , Ca^{2+} , organic carbon ($P<0.01$), and NH_4^+ ($P<0.05$). Therefore, K^+ , Ca^{2+} , NH_4^+ , organic carbon, EC in lime soil can also indirectly affect the growth of soybeans by affecting soil sucrase and alkaline phosphatase activities. On the contrary, fresh weight, dry weight, and chlorophyll content were significantly negatively correlated with NO_3^- , and PO_4^{3-} was significantly negatively correlated with fresh weight. These results further indicated that increased leaching of NO_3^- and PO_4^{3-} was not conducive to the growth of soybeans. Based on the above analysis, the addition of CA–MRB300 to lime soil can indirectly affect the growth of soybeans by increasing the contents of K^+ , Ca^{2+} and organic carbon as well as the activities of alkaline phosphatase and sucrase in lime soil.

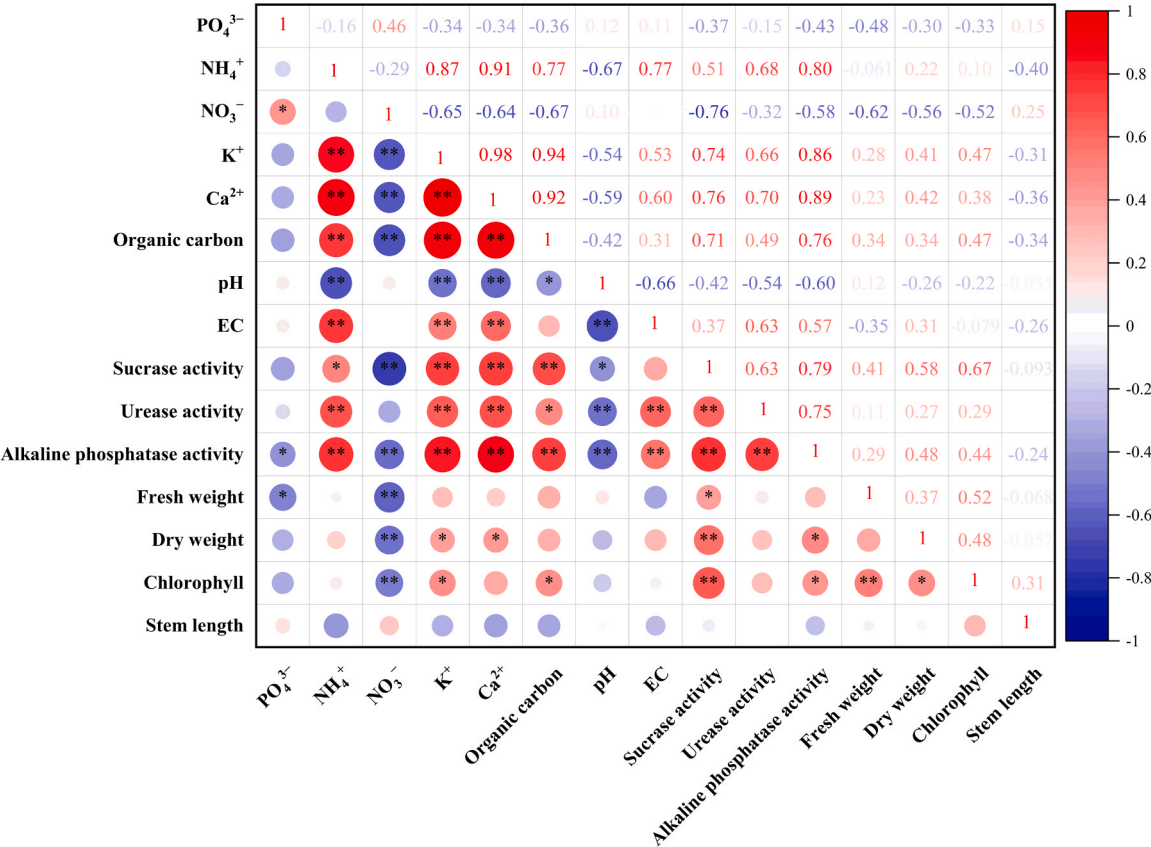


Fig. 6. Pearson's correlation heatmap between leachate properties, soil enzyme activity, and soybean growth indicators. The red and blue circles represent positive and negative correlations, respectively, and the sizes of the circles indicates the strength of the correlation.

4. Conclusions

In this study, calcium alginate–biochar composite synthesized by entrapment method was used to explore its effects of soil nutrients retention and plant growth. The results showed that the composite effectively retained PO_4^{3-} and NO_3^- in lime soil mainly through surface precipitation and physical adsorption. The composite promoted the activities of urease, sucrase, and alkaline phosphatase in lime soil. Promoted the activity of urease accelerated urea hydrolyzed into NH_4^+ . The leaching amount of K^+ was proportional to the addition ratio of the composite. Moreover, the application of the composite alone was more beneficial to the growth of soybean than the co–application of the composite with fertilizer. These findings suggest that calcium alginate–biochar composite prepared in this study can be used for the retention of PO_4^{3-} and NO_3^- in lime soil and as an alternative fertilizer for soybean growth. In the future, more large–scale field studies need to be conducted in karst areas to explore the practical application effect of the composite.

CRediT authorship contribution statement

Qianwei Feng: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Miao Chen:** Writing – review & editing, Formal analysis. **Bing Wang:** Writing – review & editing, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Xueyang Zhang:** Formal analysis. **Jian Zhang:** Supervision, Resources. **Pan Wu:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.eti.2024.103670](https://doi.org/10.1016/j.eti.2024.103670).

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